

LESSON 07 / 第七課

Tree Risk Assessment: From Targets to Residual Risk

樹木風險評估：從目標到剩餘風險

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Risk is not a synonym for defect. ANSI A300 Part 9 and the ISA BMP organise the work as targets first, then likelihood of failure of a specified part in a specified weather envelope and time frame, then consequence given occupancy. Hazard is a property of the part; risk is that property multiplied by a target. Klein and Koeser have shown that likelihood ratings are noisy across assessors. Level 1, 2 and 3 define the information collected, not a moral rank. Mitigation is a hierarchy that ends in a residual-risk sentence, not in a promise of zero. t/R and a tomogram colour are inputs to likelihood, not substitutes for the sentence.

風險不是缺陷的同義詞。ANSI A300 第 9 部分與 ISA BMP 把工作組織為：先目標，再在指定天氣包絡與時間框架內指定部位的失效可能性，然後是給定佔用下的後果。危險是部位的屬性；風險是該屬性乘上目標。Klein 與 Koeser 顯示可能性評級在評估者之間嘈雜。第 1、2、3 級定義所收集的資訊，不是道德等級。減緩是層級，止於剩餘風險句子，而不是零的承諾。t/R 與斷層顏色是可能性的輸入，不是句子的替代。

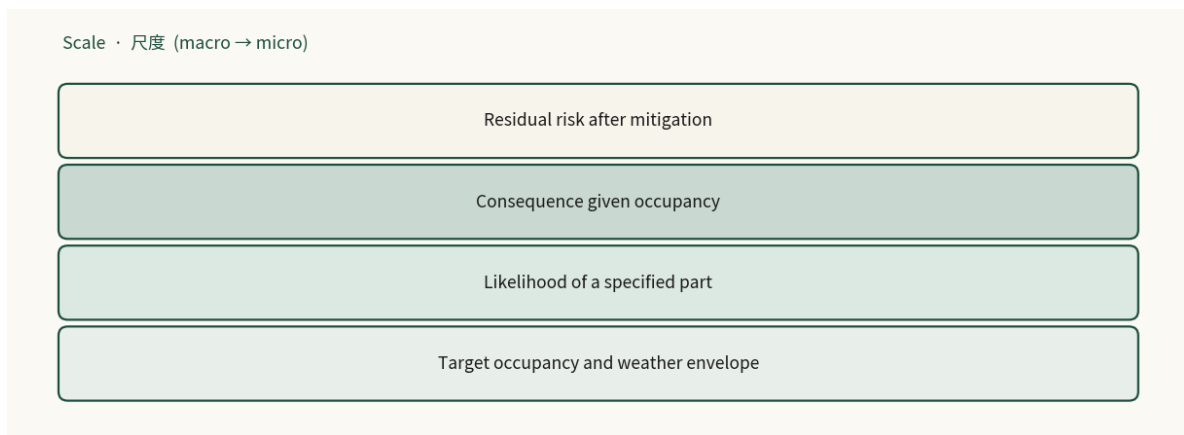


Figure 1. Nested scales from occupancy to residual risk.

圖 1. 由佔用到剩餘風險的嵌套尺度。

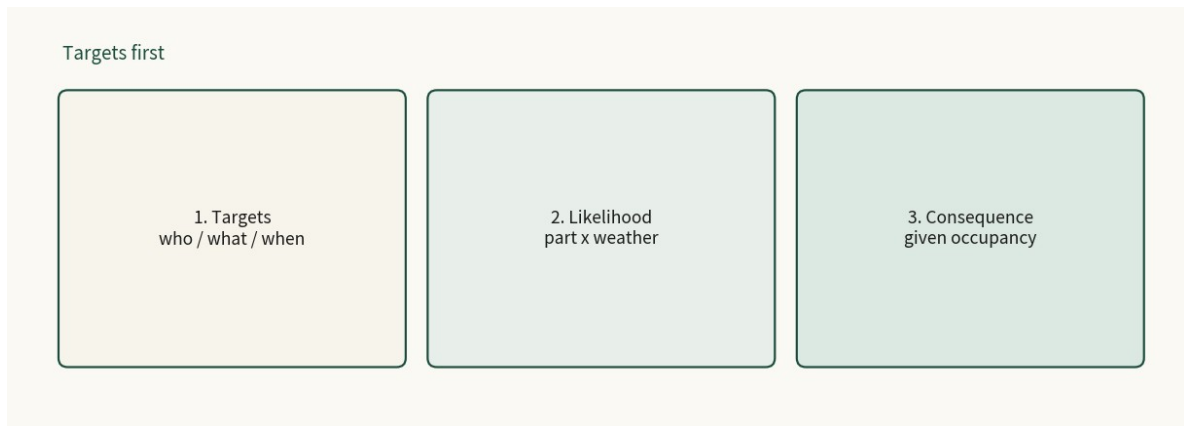


Figure 2. Targets → likelihood → consequence. Hazard is not risk.

圖2. 目標 → 可能性 → 後果。危險不是風險。

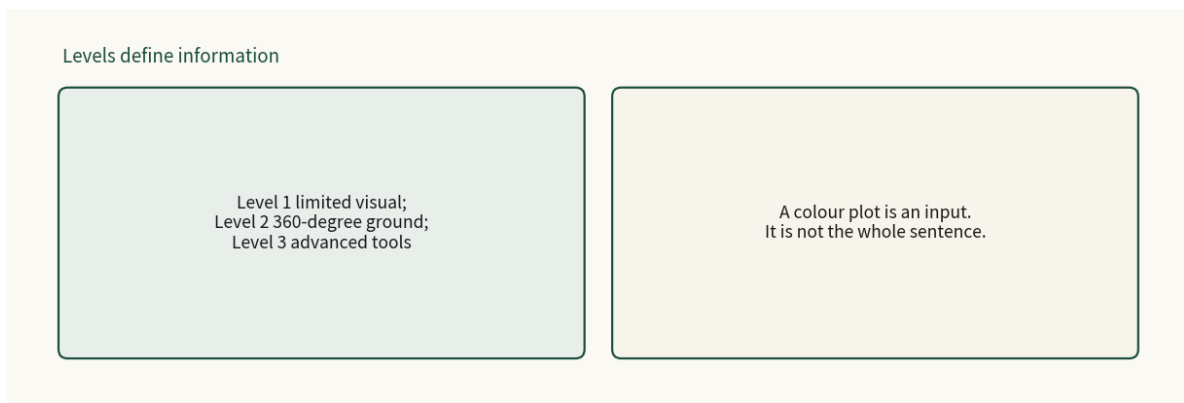


Figure 3. Assessment levels define information collected, not virtue.

圖3. 評估級別定義所收集的資訊，不是美德。

1. Scope

1. 範圍

The object is published tree-risk method. QTRA and TRAQ differ in arithmetic; they share the target-first logic reviewed here [1,2].

對象是已發表的樹木風險方法。QTRA 與 TRAQ 算術不同；它們共享此處綜述的目標優先邏輯 [1,2]。

2. Targets, likelihood, levels

2. 目標、可能性、級別

A part can only produce risk if something of value can be hit in the time frame [1,3]. Occupancy is a rate, not a yes. Weather envelope belongs with the target. Likelihood is of a specified part, not of ‘the tree’. Level 1 is a limited visual from a defined walk-by; Level 2 is a 360° ground assessment; Level 3 is advanced [3]. Koeser, Klein and colleagues have measured assessor disagreement on likelihood of failure; reproducibility is a property of the method [4,5].

只有在時間框架內有有價值之物可被擊中時，部位才能產生風險 [1,3]。佔用是速率，不是是否。天氣包絡與目標同在。可能性屬於指定部位，不是屬於「那棵樹」。第 1 級是由界定走過路徑做的有限目視；第 2 級是 360° 地面評估；第 3 級是進階 [3]。Koeser、Klein 及同事測量了評估者在失效可能性上的分歧；可重複性是方法的屬性 [4,5]。

3. Consequence, residual risk, geometry

3. 後果、剩餘風險、幾何

Consequence is given occupancy and the part that fails. Mitigation (prune, support, move the target, restrict occupancy, remove) changes one or more terms and leaves a residual-risk sentence [1,3]. 'Made safe' is not a published term unless the residual sentence is written. t/R, cavity maps and tomograms inform likelihood. Bond and Gruber argued they are not stand-alone pass/fail laws [6,7]. Using a colour plot as the whole assessment skips the target and the weather envelope.

後果是在給定佔用與失效部位之下。減緩（修剪、支撐、移動目標、限制佔用、移除）改變一個或多個項，並留下剩餘風險句子 [1,3]。除非寫出剩餘句子，「已變安全」不是已發表用語。t/R、空洞圖與斷層圖為可能性提供資訊。Bond 與 Gruber 主張它們不是獨立的通過／失敗定律 [6,7]。把顏色圖當成整個評估，等於跳過目標與天氣包絡。

4. Limits

4. 限度

Ratings are not frequencies until a defined population has been followed to failure. Most municipal systems do not have that follow-up. The honest product is a structured judgement with named uncertainty, not a fake precision to two decimals.

除非已對界定族群追隨到失效，否則評級不是頻率。多數市政系統沒有該追隨。誠實產品是帶有點名不確定性的結構化判斷，不是假到兩位小數的精確。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. Hazard and risk in this review are:

1. 本綜述中危險與風險是：

- A. Synonyms.
- A. 同義詞。
- B. Different: risk requires a target.
- B. 不同：風險需要目標。
- C. Both P50.
- C. 都是 P50。
- D. Both Wall 4.
- D. 都是第 4 壁。

2. The first term to specify is:

2. 首先要指定的項是：

- A. Paint brand.
- A. 漆品牌。
- B. Targets (who / what / when).
- B. 目標（誰／什麼／何時）。
- C. i-Tree carbon.

C. i-Tree 碳。

D. Inoculant.

D. 接種劑。

3. Likelihood is of:

3. 可能性屬於：

A. 'The tree' as a mystic whole.

A. 作為神秘整體的「那棵樹」。

B. A specified part in a specified envelope.

B. 指定包絡中的指定部位。

C. Only soil pH.

C. 只有土壤 pH。

D. Only family-level diversity.

D. 只有科級多樣性。

4. Level 1 / 2 / 3 define:

4. 第 1／2／3 級定義：

A. Moral rank.

A. 道德等級。

B. Information collected.

B. 所收集的資訊。

C. Cable sizes.

C. 鋼纜尺寸。

D. P88.

D. P88。

5. Koeser / Klein reproducibility work found likelihood ratings:

5. Koeser／Klein 可重複性工作發現可能性評級：

A. Perfectly identical across assessors.

A. 評估者之間完全相同。

B. Noisy across assessors.

B. 在評估者之間嘈雜。

C. Unnecessary.

C. 不必要。

D. Equal to t/R.

D. 等於 t/R。

6. Mitigation ends in:

6. 減緩止於：

A. A promise of zero.

A. 零的承諾。

B. A residual-risk sentence.

B. 剩餘風險句子。

C. A catalogue.

C. 一本目錄。

D. A sacred month.

D. 神聖月份。

7. t/R in this review is:

7. 本綜述中的 t/R 是：

- A. A stand-alone law.
A. 獨立定律。
- B. An input to likelihood, not the whole sentence.
B. 可能性的輸入，不是整句。
- C. A mycorrhizal index.
C. 菌根指數。
- D. A 3-30-300 rule.
D. 3-30-300 規則。

8. Occupancy is:

8. 佔用是：

- A. A yes/no essence.
A. 是否本質。
- B. A rate that varies in time.
B. 隨時間而變的速率。
- C. Wall 1.
C. 第 1 壁。
- D. A nursery system.
D. 苗圃系統。

9. 'Made safe' without a residual sentence is:

9. 沒有剩餘句子的「已變安全」是：

- A. A published term.
A. 已發表用語。
- B. Not a published term in this review.
B. 在本綜述中不是已發表用語。
- C. Choat 2012.
C. Choat 2012。
- D. A palm meristem.
D. 棕櫚分生組織。

10. QTRA and TRAQ in this review:

10. 本綜述中的 QTRA 與 TRAQ：

- A. Share no logic.
A. 不共享邏輯。
- B. Differ in arithmetic and share target-first logic.
B. 算術不同，共享目標優先邏輯。
- C. Are soil tests.
C. 是土壤檢測。
- D. Are lightning paths.
D. 是雷電路徑。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

- [1] Smiley E.T., Matheny N., Lilly S. ISA BMP Tree Risk Assessment.
- [2] Ellison M.J. QTRA method papers.
- [3] ANSI A300 Part 9. Tree Risk Assessment.
- [4] Koeser A.K. et al. likelihood-of-failure rating studies.
- [5] Klein R.W., Koeser A.K. et al. (2023). reproducibility of tree risk assessment.
- [6] Bond J. (2006). t/R geometric critique.
- [7] Gruber F. (2008). t/R cannot predict stem failure.
- [8] Dunster J.A. et al. Tree Risk Assessment Manual (ISA).

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。